Topic: Second order linear homogeneous equations

Question: Find the general solution to the second order equation.

$$y'' + 5y' + 4y = 0$$

Answer choices:

A $y(x) = c_1 e^{4x} + c_2 e^{x}$

B $y(x) = c_1 e^{4x} + c_2 e^{-x}$

C $y(x) = c_1 e^{-4x} + c_2 e^{-x}$

D $y(x) = c_1 e^{-4x} + c_2 e^{x}$

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Solution: C

The characteristic equation associated with the differential equation is

$$r^2 + 5r + 4 = 0$$

This quadratic equation is easily factorable,

$$(r + 4)(r + 1) = 0$$

which means the roots will be

$$r_1 + 4 = 0 \qquad r_2 + 1 = 0$$

$$r_1 = -4 \qquad r_2 = -1$$

These are real-numbered distinct roots, so we call them distinct real roots. Therefore, the general solution to the second order equation is

$$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

$$y(x) = c_1 e^{-4x} + c_2 e^{-x}$$

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Topic: Second order linear homogeneous equations

Question: Find the general solution to the second order equation.

$$y'' + 2y' + 11y = 0$$

Answer choices:

A $y(x) = e^{-\sqrt{10}x}(c_1 \cos(-x) + c_2 \sin(-x))$

B $y(x) = e^{\sqrt{10}x}(c_1 \cos(-x) + c_2 \sin(-x))$

C $y(x) = e^{-x}(c_1 \cos(\sqrt{10}x) + c_2 \sin(\sqrt{10}x))$

D $y(x) = e^{x}(c_1 \cos(\sqrt{10}x) + c_2 \sin(\sqrt{10}x))$

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Solution: C

The characteristic equation associated with the differential equation is

$$r^2 + 2r + 11 = 0$$

The left side doesn't easily factor, so we'll apply the quadratic formula.

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{-2 \pm \sqrt{2^2 - 4(1)(11)}}{2(1)}$$

$$r = \frac{-2 \pm \sqrt{-40}}{2}$$

Now we'll use the imaginary number to rewrite the root of the negative number.

$$r = \frac{-2 \pm \sqrt{(40)(-1)}}{2}$$

$$r = \frac{-2 \pm 2\sqrt{(10)(-1)}}{2}$$

$$r = -1 \pm \sqrt{10}i$$

These are complex-numbered roots, so we call them complex conjugate roots. If we match up $r = \alpha \pm \beta i$ with $r = -1 \pm \sqrt{10}i$, we identify $\alpha = -1$ and $\beta = \sqrt{10}$, and the general solution is therefore

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$$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

$$y(x) = e^{-x}(c_1 \cos(\sqrt{10}x) + c_2 \sin(\sqrt{10}x))$$

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Topic: Second order linear homogeneous equations

Question: If $y(0) = 2$ and $y'(0) = -5$, find the general solution to the second order equation.

$$25y'' + 30y' + 9y = 0$$

Answer choices:

A $y(x) = 2e^{-\frac{3}{5}x} - \frac{19}{5}xe^{-\frac{3}{5}x}$

B $y(x) = 2e^{-\frac{3}{5}x} - \frac{19}{5}e^{-\frac{3}{5}x}$

C $y(x) = 2e^{-\frac{3}{5}x} - \frac{5}{7}xe^{-\frac{3}{5}x}$

D $y(x) = 2e^{-\frac{3}{5}x} - \frac{5}{7}e^{-\frac{3}{5}x}$

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Solution: A

The characteristic equation associated with this differential equation is

$$25r^2 + 30r + 9 = 0$$

$$(5r + 3)(5r + 3) = 0$$

The roots are therefore

$$5r_1 + 3 = 0 \qquad 5r_2 + 3 = 0$$

$$5r_1 = -3 \qquad 5r_2 = -3$$

$$r_1 = -\frac{3}{5} \qquad r_2 = -\frac{3}{5}$$

With these equal real roots, the general solution is

$$y(x) = c_1 e^{-\frac{3}{5}x} + c_2 x e^{-\frac{3}{5}x}$$

and its derivative is

$$y(x) = -\frac{3}{5} c_1 e^{-\frac{3}{5}x} + c_2 e^{-\frac{3}{5}x} - \frac{3}{5} c_2 x e^{-\frac{3}{5}x}$$

We'll substitute the initial condition $y(0) = 2$ into $y(x)$,

$$2 = c_1 e^{-\frac{3}{5}(0)} + c_2 (0) e^{-\frac{3}{5}(0)}$$

$$2 = c_1 (1) + c_2 (0)$$

$$c_1 = 2$$

and the condition $y'(0) = -5$ into the derivative.

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$$-5 = -\frac{3}{5}c_1 e^{-\frac{3}{5}(0)} + c_2 e^{-\frac{3}{5}(0)} - \frac{3}{5}c_2(0)e^{-\frac{3}{5}(0)}$$

$$-5 = -\frac{3}{5}c_1(1) + c_2(1)$$

$$-5 = -\frac{3}{5}c_1 + c_2$$

$$-25 = -3c_1 + 5c_2$$

Substitute $c_1 = 2$ into this equation.

$$-25 = -3(2) + 5c_2$$

$$-19 = 5c_2$$

$$c_2 = -\frac{19}{5}$$

So the general solution is

$$y(x) = c_1 e^{-\frac{3}{5}x} + c_2 x e^{-\frac{3}{5}x}$$

$$y(x) = 2e^{-\frac{3}{5}x} - \frac{19}{5}xe^{-\frac{3}{5}x}$$

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